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Final Report

Title:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)
Calvert Study No.:	0835RL35.001
Testing Facility:	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447
Study Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA 92121
Date:	9 Dec 2014

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I. Table of Contents

II. List of Abbreviations

°C	Degrees Celsius
µg	Microgram
µg*hr/mL	Microgram x hour per milliliter
µg/mL	Microgram per milliliter
%	Percent
%RSD	Percent relative standard deviation
$AUC_{0\text{-}last}$	Area under the curve from time zero to the time of the last quantifiable concentration
$AUC_{0\text{-}inf}$	Area under the curve from time zero to infinity
BLQ	Below the limit of quantitation
C_o	Extrapolated concentration at time zero
C_{last}	Last quantifiable concentration
Cl	Apparent plasma clearance
C_{max}	Maximum observed concentration; for intravenous bolus dosing the C_{max} is considered to occur at the first sampling time after dose administration
Conc	Concentration
g	Gram
hr	Hour
hr:min:sec	Hours:minutes:seconds
IACUC	Institutional animal care and use committee
K_2EDTA	Di-potassium ethylenediaminetetraacetic acid
kg	Kilogram
M	Male
mg	Milligram
mg/kg	Milliligram per kilogram

mg/mL	Milligram per milliliter
min	Minute
mL	Milliliter
mL/kg	Milliliter per kilogram
NA	Not applicable
ND	Not determined
No.	Number
NRC	National Research Council
rpm	Revolutions per minute
SD	Standard deviation
sec	Second
SOP	Standard Operating Procedure
$T_{1/2}$	Apparent plasma elimination phase half-life
T_{last}	Time of last quantifiable concentration
T_{max}	Time when the maximum concentration was observed; for intravenous bolus dosing the C_{max} is considered to occur at the first sampling time after dose administration
V_c	Apparent volume of distribution from the central compartment
V_{ss}	Apparent volume of distribution at steady-state
Vol	Volume

III. Signatures

Calvert Study No.: 0835RL35.001

Title: A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)

A. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained.

Steven D. Sloneker — 9 Dec 2014

Steven Sloneker, B.S.
Study Director
Calvert Laboratories, Inc.

Date

B. Signature of Other Responsible Personnel

Scientific Oversight and Report Review

9 DEC 2014

Scientific Management
Calvert Laboratories, Inc.

Date

IV. Summary

A. Title

A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)

B. Objective

The purpose of this study was to evaluate the pharmacokinetics of LT3114 in male Sprague Dawley rats after a single intravenous dose of LT3114 at two different dose levels.

C. Methods

Two groups of four male Sprague-Dawley rats were administered LT3114 as a single manual slow bolus intravenous injection over a period of ~3 minutes into a lateral tail vein as per the targeted protocol specified parameters below.

Group	# of Animals and Sex	Route of Administration	Dose Conc. (mg/mL)	Dose Volume (mL/kg)	Dose Level (mg/kg)
1	4 males	IV	2	5	10
2	4 males	IV	10	5	50

Serial blood samples were obtained from each animal at 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose. Derived plasma samples were stored frozen at -70°C until shipment to Burleson Research Technologies (Morrisville, NC). LT3114 rat plasma concentration-time results were forwarded to Calvert for pharmacokinetic analysis.

D. Results

Intravenous administration via a 3-minute slow bolus hand-push injection was conducted in all eight animals without incident. All blood samples were harvested at their targeted time of collection. All animals appeared normal during dosing and after dosing with the exception of one animal. Rat# 1350 was observed with soft feces at 24, 48, 72, and 120 hours post-dose. Mean assayed LT3114 concentrations for the dosing formulations were 1.95 ± 0.255 mg/mL (%CV = 13.1) and 13.9 ± 0.81 mg/mL (%CV = 5.8).

Nominal mean ± standard deviation dosing details are summarized in the following table.

Group #	Animals and Sex	Route of Adminstration	Nominal Dose Conc. (mg/mL)	Nominal Dose Volume (g/kg)	Nominal Dose Level (mg/kg)
1	1343-1346 males	IV	1.926	5.10 ± 0.114	9.82 ± 0.220
2	1347-1350 males	IV	9.951	5.24 ± 0.0852	52.1 ± 0.85

Mean ± SD; N=4

The nominal dose concentration was based on the actual formulation procedure. For Group 1, 1.2 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 18.8 mL of vehicle. For Group 2, 6.2 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 13.8 mL of vehicle. The nominal dose volume was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle. The nominal dose level was calculated as the nominal dose concentration multiplied by the nominal dose volume.

Using noncompartmental analysis and WinNonlin Version 5.3 software, pharmacokinetic parameters were calculated for each individual animal using the animal's LT3114 plasma concentration-time data. Individual animal nominal dose level based on the animal's body weight and nominal dose concentration and nominal dose volume was used in the pharmacokinetic analysis. Mean ± standard deviation pharmacokinetic parameter values were then calculated from the four animals in each dose group. A summary table of the mean ± standard deviation pharmacokinetic parameters is presented below.

Group	Dose (mg/kg)	C_0 (µg/mL)	$AUC_{0\text{-last}}$ (µg*hr/mL)	$AUC_{0\text{-inf}}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	9.82 ±0.220	174 ±27.2	21206 ±1701.6	25374 ±2520.2	190 ±34.8	0.390 ±0.05122	98.3 ±10.43	57.9 ±12.05
2	52.1 ±0.85	1104 ±72.3	108988 ±7438.7	139012 ±9042.6	233 ±33.2	0.376 ±0.02625	115 ±9.68	47.3 ±2.20

Dose and pharmacokinetic parameters presented as Mean ± SD; N=4

Based on C_0, $AUC_{0\text{-last}}$, $AUC_{0\text{-inf}}$ and concentration-time profiles, exposure to LT3114 was LT3114 dose-dependent and this appeared to be dose-proportional. Mean C_0 was 174 ± 27.2 µg/mL and mean $AUC_{0\text{-inf}}$ was 25374 ± 2520.2 µg*hr/mL for the 9.82 mg/kg dose group. Mean C_0 was 1104 ± 72.3 µg/mL and mean $AUC_{0\text{-}inf}$ was 139012 ± 9042.6 µg*hr/mL for the 52.1 mg/kg dose group. Mean terminal plasma $T_{1/2}$ was 190 ± 34.8 hours and 233 ± 33.2 hours following intravenous administration of 9.82 mg/kg and 52.1 mg/kg, respectively. In all profiles, T_{last} was observed at the last blood sampling timepoint of 480 hours. Systemic clearance and apparent volume of distribution values were similar per dose level. Following 9.82 mg/kg, mean Cl was 0.390 ± 0.05122 mL/hr/kg, mean V_{ss} was 98.3 ± 10.43 mL/kg, and mean V_c was 57.9 ± 12.05 mL/kg. Following 52.1 mg/kg, mean Cl was 0.376 ± 0.02625 mL/hr/kg, mean V_{ss} was 115 ± 9.68 mL/kg, and mean V_c was 47.3 ± 2.20 mL/kg.

V. General Information

A. Key Study Dates

Study Initiation Date:	27 Jun 2014
Animal Receipt Date:	17 Jul 2014
Experimental Start Date (in-life):	21 Jul 2014
Experimental Completion Date (in-life):	10 Aug 2014

B. Responsible Personnel at Testing Facility

Study Director:	Steven Sloneker, B.S.
Primary Technician:	Anthony Colazzo, B.S., LAT and Carrie Olechna, A.S.

C. Contributing Scientist

Study Monitor:	Marlene W. Modi, Ph.D. MWM Consulting Group Inc. for Lpath Inc. 14 Joseph Court Monmouth Junction, NJ 08852
Sponsor Representative:	Gary Woodnutt, Ph.D. Senior Vice President Research, Lpath, Inc. 4025 Sorrento Valley Blvd. San Diego, CA 92121
Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis:	Victor J. Johnson, Ph.D. Burleson Research Technologies 120 First Flight Lane Morrisville, NC 27560

D. Objective

The purpose of this study was to evaluate the pharmacokinetics of LT3114 in male Sprague Dawley rats after a single intravenous dose of LT3114 at two different dose levels.

VI. Materials and Methods

A. Test Article

1. Test Article

Identification:	LT3114
Lot/Batch No. :	103948
Physical Description:	Clear colorless liquid
Storage Conditions:	Stored refrigerated (2-8°C) and protected from light.

B. Dose Preparations

Before dose preparation, 2 mL of a 10% polysorbate 80 solution (provided by the Sponsor; Batch# 104012-001, MM#306191), was added aseptically to each liter of formulation buffer (also supplied by the Sponsor and in sterile containers; Batch# 104011-001), and mixed gently to ensure distribution throughout the volume of buffer. The Calvert batch number for this vehicle in this study was 30 June 2014 and the vehicle was stored at 2 to 8 °C until use. The Sponsor provided the test article already formulated at 32.1 mg/mL in sterile containers (a Certificate of Analysis for this formulated material is included in Appendix I). Vehicle (24.3 mM citrate, 51.4 nM sodium phosphate, and 300 mM D-trehalose containing the 2 mL of a 10% polysorbate 80 solution and test article were removed from refrigeration and allowed to warm to room temperature for 33 minutes. All formulation procedures were conducted in a cleaned laminar flow hood using aseptic techniques on 21 Jul 2014.

<u>Targeted 2 mg/mL LT3114 Preparation:</u> 1.2 mL of the nominal 32.1 mg/mL LT3114 test article solution was mixed with 18.8 mL of the vehicle. The contents of the formulation vial were gently swirled until the preparation appeared homogeneous. Triplicate 1.0 mL samples were collected and stored at 2 to 8 °C. pH was measured at 5. The nominal LT3114 concentration was 1.926 mg/mL.

<u>Targeted 10 mg/mL LT3114 Preparation:</u> 6.2 mL of the nominal 32.1 mg/mL LT3114 test article solution was mixed with 13.8 mL of the vehicle. The contents of the formulation vial were gently swirled until the preparation appeared homogeneous. Triplicate 1.0 mL samples were collected and stored at 2 to 8 °C. pH was measured at 5. The nominal LT3114 concentration was 9.951 mg/mL.

C. Dose Formulation Samples

Dose formulation samples were shipped refrigerated on chill packs on 11 Aug 2014 to and analytical evaluation was done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. A cover letter, sample transmittal form, and sample log was sent along with the samples.

D. Accountability and Disposition

Unused test article will be returned to the Sponsor at the termination of this study or, if requested by the Sponsor, retained for use on related future studies.

E. Test System (Animals and Animal Care)

1. Description

Species:	Rat
Stock:	Sprague Dawley Crl:CD(SD)
Total Number:	8; each animal was surgically implanted with a catheter in a jugular vein using polyurethane tubing and a heparin/50% dextrose solution as the lumen lock solution.
Gender:	Male
Age Range:	9 weeks at start of dosing; records of dates of birth for animals used in this study will be retained in the Calvert archives.
Body Weight Range:	309 to 331 g
Animal Source:	Charles River Laboratories, Raleigh, NC
Experimental History:	Purpose-bred and experimentally naïve at the outset of the study
Identification:	Ear tag and cage card

2. Rationale for Choice of Species and Number of Animals

The Sprague Dawley rat was chosen for this study as it is a species that is used for non-clinical toxicity and pharmacokinetic evaluations and satisfies the regulatory requirement for non-clinical safety studies in rodents (1).

Previous research has been conducted with the test article in rats. The test article, a human anti-LPA antibody (LT3114) and the murine parental molecule have both been evaluated for efficacy in a variety of rat and mouse animal models of disease. In models using Sprague Dawley rats efficacy has been examined in the STZ induced model of diabetic neuropathy, a chemotherapeutic- induced model of peripheral neuropathy and in models of acute inflammatory pain (carrageen or Freunds adjuvant induced) with doses (intravenous or subcutaneous) ranging from 10 to 40 mg/kg.

In addition single and multiple doses pharmacokinetic studies have been conducted with LT3114 in murine models at dose levels ranging from 10 to 30 mg/kg. Four animals per dose group allowed the use of descriptive statistical analysis of the data.

3. *Husbandry*

Housing:	Animals were individually housed in plastic totes with bedding upon receipt in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals were kept was documented in the study records. No other species were kept in the same room.
Lighting:	12 hours light/12 hours dark except when room lights were turned on during the dark cycle due to in-life procedures.
Room Temperature:	17 to 24°C
Relative Humidity:	30 to 92%
Food:	All animals had access to Harlan Teklad Rodent Diet (certified) *ad libitum*. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants were known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed

	by the manufacturer, records of which are maintained in the Calvert archives.
Water:	Water was available *ad libitum* to each animal. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants were known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives.
Acclimation:	Study animals were acclimated to their laboratory environment for 4 days prior to dosing.

4. Prestudy Health Screen and Selection Criteria

All animals received for this study were assessed as to their general health by a member of the veterinary staff or other authorized personnel. During the acclimation period, each animal was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use were assigned to this study.

5. Assignment to Study Groups

Animals were manually assigned to the study based on general health, body weight and catheter patency. Animals were assigned a permanent identification number and then ear tagged.

6. Humane Care of Animals

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert IACUC approved the study protocol prior to finalization.

F. Test Article Administration

1. Treatment and Dosing Details

Group	# of Animals and Sex	Route of Administration	Dose Conc. (mg/mL)	Dose Volume (mL/kg)	Dose Level (mg/kg)
1	4 males	IV	2	5	10
2	4 males	IV	10	5	50

The dose concentration, dose volume, and dose level values in the table above represent the protocol targeted values.

2. Dosing

Route: Intravenous

Frequency: Single intravenous injection

Procedures: The intravenous formulation was administered as a manual slow bolus injection over a period of approximately 3 minutes into a lateral tail vein.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans. Dose levels were selected by the Sponsor. The anticipated lowest pharmacologically active dose is 30 mg/kg.

G. In-Life Observations and Measurements

1. Clinical Observations

Animals were observed at least once daily during acclimation and at all times of blood sampling post-dose.

2. Body Weight

Body weight was determined on the day of test article administration just prior to dosing.

H. Sample Collection

1. Blood Sampling

Twelve serial blood samples (~0.25 mL each) were obtained from each animal via the jugular vein catheter at the following timepoints: 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose. Collected blood samples were transferred into pre-labeled tubes containing K_2 EDTA anticoagulant. Just before each blood sampling timepoint, blood was withdrawn (~43 µL) and discarded. A blood sample (approximately 0.25 mL) was obtained from the jugular vein catheter using a syringe with blunted needle and then the blood sample was immediately transferred into a blood collection tube. Following blood sample collection at 0.5, 3, and 6 hours post-dose, the catheter was flushed with sterile saline. Following blood sample collection at 12, 24, 48, 72, 120, 180, 240, and 360 hours post-dose, the catheter was flushed with sterile saline and then locked with a locking solution (taurolidine-citrate solution). The flush volume was slightly larger than the catheter tubing void volume. Collected blood samples were gently inverted several times and stored on chill packs until centrifugation within 10 minutes of collection.

2. *Plasma Harvesting*

Blood samples were centrifuged at 3000 rpm at +4 °C for 10 minutes. Each derived plasma sample was transferred into a prelabeled plastic tube labeled with the study number, species, animal number, dose level/route, date of collection, time of collection, and sample type. Derived plasma samples were stored frozen at approximately -70 °C until shipment to Burleson Research Technologies.

I. Terminal Procedures

1. *Termination*

Animals were euthanized by CO_2 asphyxiation following their final blood sampling. No gross necropsy was performed.

J. Sample Shipment

The plasma samples were shipped frozen on dry ice via overnight courier on 11 Aug 2014 to Burleson Research Technologies, Attention: Victor J. Johnson, Ph.D., 120 First Flight Lane, Morrisville, NC 275601. A cover letter, sample transmittal form, and sample log was sent along with the samples.

VII. Records and Reports

A. Data Analysis and Statistical Evaluation

Data analysis by Calvert consisted of descriptive statistical analysis of animal body weights and dosing details. Animal observations and blood sampling times (actual and elapsed) were documented. Analytical analysis of the dosing formulations was the responsibility of Burleson Research Technologies. Bioanalysis of the plasma samples was the responsibility of Burleson Research Technologies (BRT). Pharmacokinetic analysis of the plasma sample assay results was the responsibility of Calvert. Individual animal concentration-time data and averaged concentration-time data per dose level were used and Office Excel 2010 (Microsoft Corp., Redmond, WA) was used for graphing and figures. Any plasma value reported as BLQ was assigned a value of zero. Using noncompartmental analysis and WinNonlin Version 5.3 software, individual pharmacokinetic parameters were calculated. These included $AUC_{0\text{-inf}}$, $AUC_{0\text{-last}}$, C_{max}, Cl, C_{last}, C_0, T_{last}, T_{max}, $T_{1/2}$, and V_{ss}. $T_{1/2}$ was determined from regression analysis of at least the last three consecutive timepoints with quantifiable LT3114 concentration. V_c was calculated by Dose / C_0. Non-truncated values were used in the calculation of all pharmacokinetic parameters and for all descriptive statistical analysis.

B. Storage of Records

Test article preparation, test article tracking, in-life data, protocol, protocol amendments, and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. Approximately 2 years after finalization of the report, the Sponsor will be contacted to determine final disposition of all study materials.

VIII. Results

Individual and mean animal body weights and dosing details are summarized in Table 1. Animal observations and blood sampling times (actual and elapsed) are summarized in Appendix III. Individual animal LT3114 concentration-time data are presented in Appendix IV. A BRT analytical/bioanalytical subreport entitled "Determination of LT3114 in Dose Formulation Samples and Rat Plasma Pharmacokinetic Samples" is presented in Appendix V. Individual and mean ± SD (%RSD) animal LT3114 concentration-time data are summarized in Table 2.

Individual LT3114 plasma concentration-time data plotted as a function of time are presented in Figure 1. Mean ± SD LT3114 plasma concentration-time data per dose level are plotted as a function of time in Figure 2. Individual and mean ± SD (%RSD) LT3114 plasma pharmacokinetic parameters per dose level are summarized in Table 3.

Intravenous administration via a 3-minute slow bolus hand-push injection was conducted in all eight animals without incident. All blood samples were harvested at their targeted time of collection. All animals appeared normal during dosing and after dosing with the exception of one animal. Rat# 1350 was observed with soft feces at 24, 48, 72, and 120 hours post-dose. Mean assayed LT3114 concentrations for the dosing formulations were 1.95 ± 0.255 mg/mL with triplicate middle sample LT3114 measured concentrations of 1.96, 2.20, and 1.69 mg/mL (%CV = 13.1) and 13.9 ± 0.81 mg/mL with triplicate middle sample LT3114 measured concentrations of 12.95, 14.40, and 14.30 mg/mL (%CV = 5.8). Assayed LT3114 concentration values were not used in the pharmacokinetic analysis.

Nominal mean ± standard deviation dosing details are summarized in the following table.

Group #	Animals and Sex	Route of Administration	Nominal Dose Conc. (mg/mL)	Nominal Dose Volume (g/kg)	Nominal Dose Level (mg/kg)
1	1343-1346 males	IV	1.926	5.10 ± 0.114	9.82 ± 0.220
2	1347-1350 males	IV	9.951	5.24 ± 0.0852	52.1 ± 0.85

Mean ± SD; N=4

The nominal dose concentration was based on the actual formulation procedure. For Group 1, 1.2 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 18.8 mL of vehicle. For Group 2, 6.2 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 13.8 mL of vehicle. The nominal dose volume was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle. The nominal dose level was calculated as the nominal dose concentration multiplied by the nominal dose volume.

Using noncompartmental analysis and WinNonlin Version 5.3 software, pharmacokinetic parameters were calculated for each individual animal using the animal's LT3114 plasma concentration-time data. Individual animal nominal dose level based on the animal's body weight and nominal dose concentration and nominal dose volume was used in the pharmacokinetic analysis. Mean ± standard deviation pharmacokinetic parameter values were then calculated from the four animals in each dose group. A summary table of the mean ± standard deviation pharmacokinetic parameters is presented below.

Group	Dose (mg/kg)	C_0 (µg/mL)	$AUC_{0\text{-last}}$ (µg*hr/mL)	$AUC_{0\text{-inf}}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	9.82 ±0.220	174 ±27.2	21206 ±1701.6	25374 ±2520.2	190 ±34.8	0.390 ±0.05122	98.3 ±10.43	57.9 ±12.05
2	52.1 ±0.85	1104 ±72.3	108988 ±7438.7	139012 ±9042.6	233 ±33.2	0.376 ±0.02625	115 ±9.68	47.3 ±2.20

Dose and pharmacokinetic parameters presented as Mean ± SD; N=4

Based on C_0, $AUC_{0\text{-last}}$, $AUC_{0\text{-inf}}$ and concentration-time profiles, exposure to LT3114 was LT3114 dose-dependent and this appeared to be dose-proportional. Mean C_0 was 174 ± 27.2 µg/mL and mean $AUC_{0\text{-inf}}$ was 25374 ± 2520.2 µg*hr/mL for the 9.82 mg/kg dose group. Mean C_0 was 1104 ± 72.3 µg/mL and mean $AUC_{0\text{-inf}}$ was 139012 ± 9042.6 µg*hr/mL for the 52.1 mg/kg dose group. Mean terminal plasma $T_{1/2}$ was 190 ± 34.8 hours and 233 ± 33.2 hours following intravenous administration of 9.82 mg/kg and 52.1 mg/kg, respectively. In all profiles, T_{last} was observed at the last blood sampling timepoint of 480 hours. Systemic clearance and apparent volume of distribution values were similar per dose level. Following 9.82 mg/kg, mean Cl was 0.390 ± 0.05122 mL/hr/kg, mean V_{ss} was 98.3 ± 10.43 mL/kg, and mean V_c was 57.9 ± 12.05 mL/kg. Following 52.1 mg/kg, mean Cl was 0.376 ± 0.02625 mL/hr/kg, mean V_{ss} was 115 ± 9.68 mL/kg, and mean V_c was 47.3 ± 2.20 mL/kg.

IX. Tables

A. Table 1 - Individual and Mean Animal Body Weights and Dosing Details

Rat #	Sex	Route	Date	Body Weight (kg)	Syringe Weights (g)			Nominal Dose Volume		Nominal Dose Level	
					Loaded	Unloaded	Net	(mL)	(g/kg)[a]	(mg)[b]	(mg/kg)[c]
1343	M	IV	21-Jul-14	0.324	7.4840	5.7779	1.7061	1.7	5.27	3.29	10.1
1344	M	IV	21-Jul-14	0.309	7.3703	5.8055	1.5648	1.6	5.06	3.01	9.75
1345	M	IV	21-Jul-14	0.329	7.4480	5.7979	1.6501	1.7	5.02	3.18	9.66
1346	M	IV	21-Jul-14	0.329	7.4473	5.7881	1.6592	1.7	5.04	3.20	9.71
Group 1			Mean	0.323	7.4374	5.7924	1.6451	1.6	5.10	3.17	9.82
			SD	0.009465	0.047904	0.011980	0.058858	0.059	0.114	0.113	0.220
			%RSD	2.9	0.6	0.2	3.6	3.6	2.2	3.6	2.2

Rat #	Sex	Route	Date	Body Weight (kg)	Syringe Weights (g)			Nominal Dose Volume		Nominal Dose Level	
					Loaded	Unloaded	Net	(mL)	(g/kg)[a]	(mg)[b]	(mg/kg)[c]
1347	M	IV	21-Jul-14	0.331	7.4750	5.7558	1.7192	1.7	5.19	17.11	51.7
1348	M	IV	21-Jul-14	0.313	7.4638	5.7856	1.6782	1.7	5.36	16.70	53.4
1349	M	IV	21-Jul-14	0.319	7.4727	5.8063	1.6664	1.7	5.22	16.58	52.0
1350	M	IV	21-Jul-14	0.323	7.4074	5.7368	1.6706	1.7	5.17	16.62	51.5
Group 2			Mean	0.322	7.4547	5.7711	1.6836	1.7	5.24	16.75	52.1
			SD	0.007550	0.031918	0.030875	0.024231	0.024	0.0852	0.241	0.85
			%RSD	2.3	0.4	0.5	1.4	1.4	1.6	1.4	1.6

[a]The nominal dose volume (g/kg) was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle.
[b]The nominal amount (mg) administered was calculated as the product of the net syringe weight and the nominal dose concentration (1.926 mg/mL for Group 1 and 9.951 mg/mL for Group 2)
[c]The nominal dose level (mg/kg) was calculated as the nominal dose concentration multiplied by the nominal dose volume.

B. Table 2 - Individual and Mean LT3114 Plasma Concentration-Time Data Following a Single Intravenous Dose of LT3114 to Male Sprague-Dawley Rats

Group #	Rat #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (µg/mL)	Mean Conc. (µg/mL)	SD	%RSD
1	1343	M	10.1	0.5	132.8			
1	1344	M	9.75	0.5	182.0			
1	1345	M	9.66	0.5	190.2			
1	1346	M	9.71	0.5	185.5	172.6	26.76	15.5
1	1343	M	10.1	3	208.5			
1	1344	M	9.75	3	172.1			
1	1345	M	9.66	3	185.5			
1	1346	M	9.71	3	190.8	189.2	15.07	8.0
1	1343	M	10.1	6	170.7			
1	1344	M	9.75	6	163.8			
1	1345	M	9.66	6	176.3			
1	1346	M	9.71	6	157.5	167.1	8.18	4.9
1	1343	M	10.1	12	121.3			
1	1344	M	9.75	12	139.2			
1	1345	M	9.66	12	133.8			
1	1346	M	9.71	12	126.4	130.2	7.91	6.1
1	1343	M	10.1	24	113.5			
1	1344	M	9.75	24	124.9			
1	1345	M	9.66	24	110.7			
1	1346	M	9.71	24	89.0	109.5	15.00	13.7
1	1343	M	10.1	48	73.2			
1	1344	M	9.75	48	77.5			
1	1345	M	9.66	48	76.6			
1	1346	M	9.71	48	76.6	76.0	1.90	2.5
1	1343	M	10.1	72	69.8			
1	1344	M	9.75	72	68.1			
1	1345	M	9.66	72	67.2			
1	1346	M	9.71	72	65.9	67.8	1.64	2.4
1	1343	M	10.1	120	50.5			
1	1344	M	9.75	120	52.4			
1	1345	M	9.66	120	47.9			
1	1346	M	9.71	120	53.5	51.1	2.45	4.8
1	1343	M	10.1	180	36.1			
1	1344	M	9.75	180	42.1			
1	1345	M	9.66	180	46.4			
1	1346	M	9.71	180	40.6	41.3	4.25	10.3
1	1343	M	10.1	240	26.1			
1	1344	M	9.75	240	43.2			
1	1345	M	9.66	240	43.7			
1	1346	M	9.71	240	35.0	37.0	8.29	22.4

B. Table 2 (continued) - Individual and Mean LT3114 Plasma Concentration-Time Data Following a Single Intravenous Dose of LT3114 to Male Sprague-Dawley Rats

Group #	Rat #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (µg/mL)	Mean Conc. (µg/mL)	SD	%RSD
1	1343	M	10.1	360	17.5			
1	1344	M	9.75	360	26.3			
1	1345	M	9.66	360	24.2			
1	1346	M	9.71	360	23.5	22.9	3.78	16.5
1	1343	M	10.1	480	10.6			
1	1344	M	9.75	480	16.7			
1	1345	M	9.66	480	16.2			
1	1346	M	9.71	480	17.5	15.3	3.15	20.6
2	1347	M	51.7	0.5	1048.4			
2	1348	M	53.4	0.5	1165.8			
2	1349	M	52.0	0.5	1037.6			
2	1350	M	51.5	0.5	1026.9	1069.7	64.68	6.0
2	1347	M	51.7	3	930.9			
2	1348	M	53.4	3	966.4			
2	1349	M	52.0	3	832.1			
2	1350	M	51.5	3	929.5	914.7	57.67	6.3
2	1347	M	51.7	6	832.1			
2	1348	M	53.4	6	778.4			
2	1349	M	52.0	6	779.6			
2	1350	M	51.5	6	827.9	804.5	29.50	3.7
2	1347	M	51.7	12	626.3			
2	1348	M	53.4	12	640.2			
2	1349	M	52.0	12	672.2			
2	1350	M	51.5	12	663.2	650.5	21.01	3.2
2	1347	M	51.7	24	515.6			
2	1348	M	53.4	24	541.7			
2	1349	M	52.0	24	485.2			
2	1350	M	51.5	24	529.7	518.1	24.36	4.7
2	1347	M	51.7	48	478.0			
2	1348	M	53.4	48	432.3			
2	1349	M	52.0	48	424.4			
2	1350	M	51.5	48	454.5	447.3	24.11	5.4
2	1347	M	51.7	72	274.0			
2	1348	M	53.4	72	306.6			
2	1349	M	52.0	72	274.0			
2	1350	M	51.5	72	312.3	291.7	20.60	7.1
2	1347	M	51.7	120	224.2			
2	1348	M	53.4	120	224.2			
2	1349	M	52.0	120	247.2			
2	1350	M	51.5	120	282.3	244.5	27.45	11.2

B. Table 2 (continued) - Individual and Mean LT3114 Plasma Concentration-Time Data Following a Single Intravenous Dose of LT3114 to Male Sprague-Dawley Rats

Group #	Rat #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (µg/mL)	Mean Conc. (µg/mL)	SD	%RSD
2	1347	M	51.7	180	254.1			
2	1348	M	53.4	180	281.1			
2	1349	M	52.0	180	190.8			
2	1350	M	51.5	180	224.8	237.7	38.81	16.3
2	1347	M	51.7	240	170.7			
2	1348	M	53.4	240	225.0			
2	1349	M	52.0	240	151.6			
2	1350	M	51.5	240	159.6	176.7	33.12	18.7
2	1347	M	51.7	360	133.3			
2	1348	M	53.4	360	138.8			
2	1349	M	52.0	360	112.4			
2	1350	M	51.5	360	143.4	132.0	13.69	10.4
2	1347	M	51.7	480	92.5			
2	1348	M	53.4	480	96.0			
2	1349	M	52.0	480	77.5			
2	1350	M	51.5	480	90.6	89.2	8.08	9.1

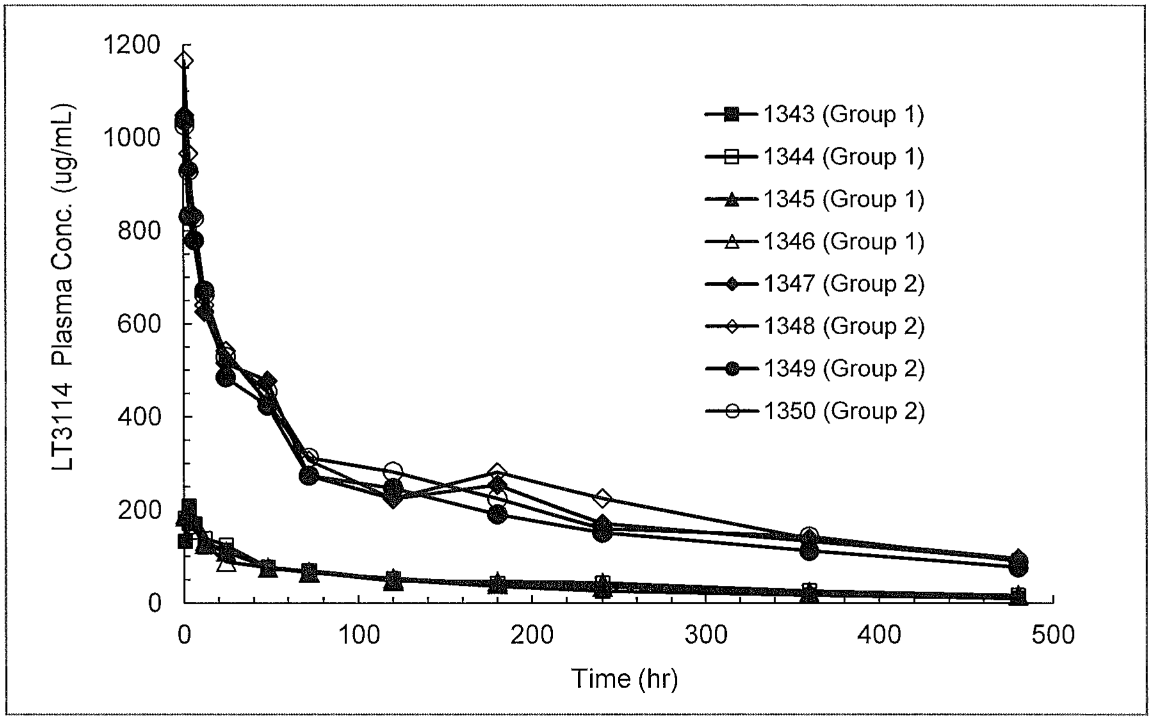
C. Table 3 - Individual and Mean Pharmacokinetic Parameters in Plasma Following a Single Intravenous Dose of LT3114 to Male Sprague-Dawley Rats

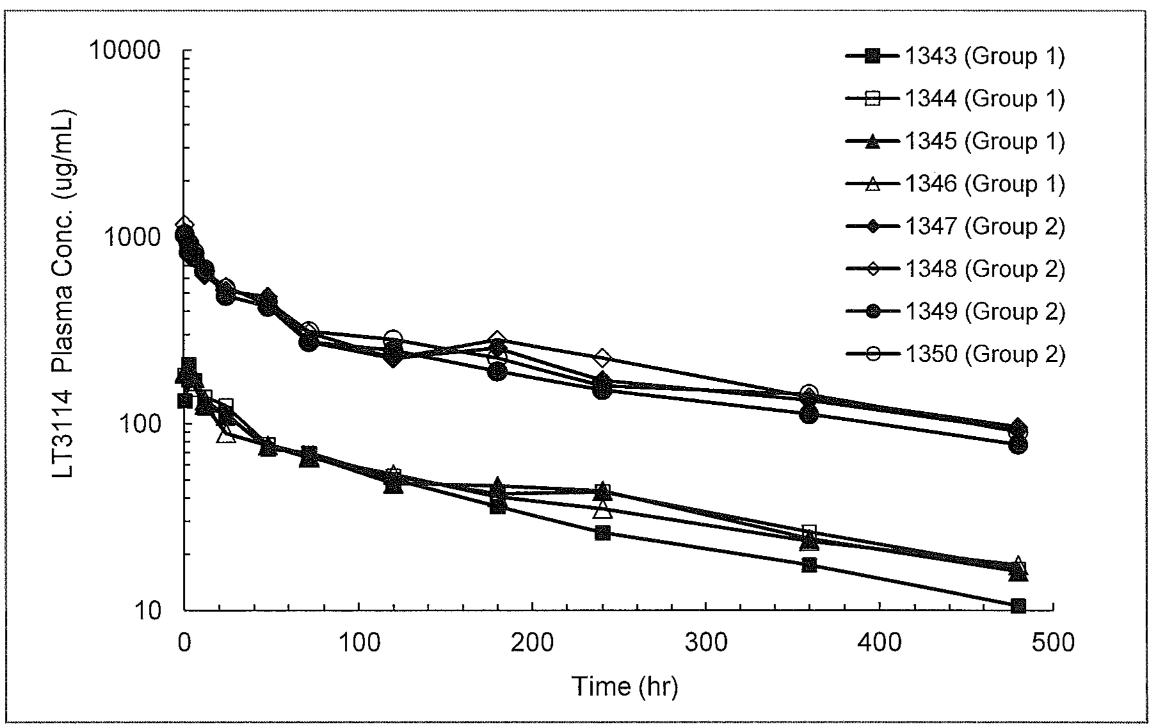
Group #	Rat #	Dose (mg/kg)	C_0 (µg/mL)	C_{max} (µg/mL)	T_{max} (hr)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	1343	10.1	133	209	3	10.6	480	18956	21639	175	0.467	100	75.9
1	1344	9.75	184	182	0.5	16.7	480	22744	26931	175	0.362	90.8	53.0
1	1345	9.66	191	190	0.5	16.2	480	22271	26067	168	0.371	90.1	50.6
1	1346	9.71	186	191	3	17.5	480	20853	26857	242	0.362	112	52.2
	Mean	9.82	174	193	1.8	15.3	480	21206	25374	190	0.390	98.3	57.9
	SD	0.220	27.2	11.4	1.44	3.15	0.0	1701.6	2520.2	34.8	0.05122	10.43	12.05
	%RSD	2.0	15.7	5.9	82.5	20.6	0.0	8.0	9.9	18.3	13.1	10.6	20.8

Group #	Rat #	Dose (mg/kg)	C_0 (µg/mL)	C_{max} (µg/mL)	T_{max} (hr)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
2	1347	51.7	1074	1048	0.5	92.5	480	108661	145614	272	0.355	122	48.1
2	1348	53.4	1210	1166	0.5	96.0	480	116776	142653	191	0.374	103	44.1
2	1349	52.0	1084	1038	0.5	77.5	480	99036	125644	237	0.414	123	48.0
2	1350	51.5	1048	1027	0.5	90.6	480	111478	142136	233	0.362	110	49.1
	Mean	52.1	1104	1070	0.5	89.2	480	108988	139012	233	0.376	115	47.3
	SD	0.85	72.3	64.7	0.00	8.08	0.0	7438.7	9042.6	33.2	0.02625	9.68	2.20
	%RSD	1.6	6.5	6.1	0.0	9.1	0.0	6.8	6.5	14.2	7.0	8.5	4.7

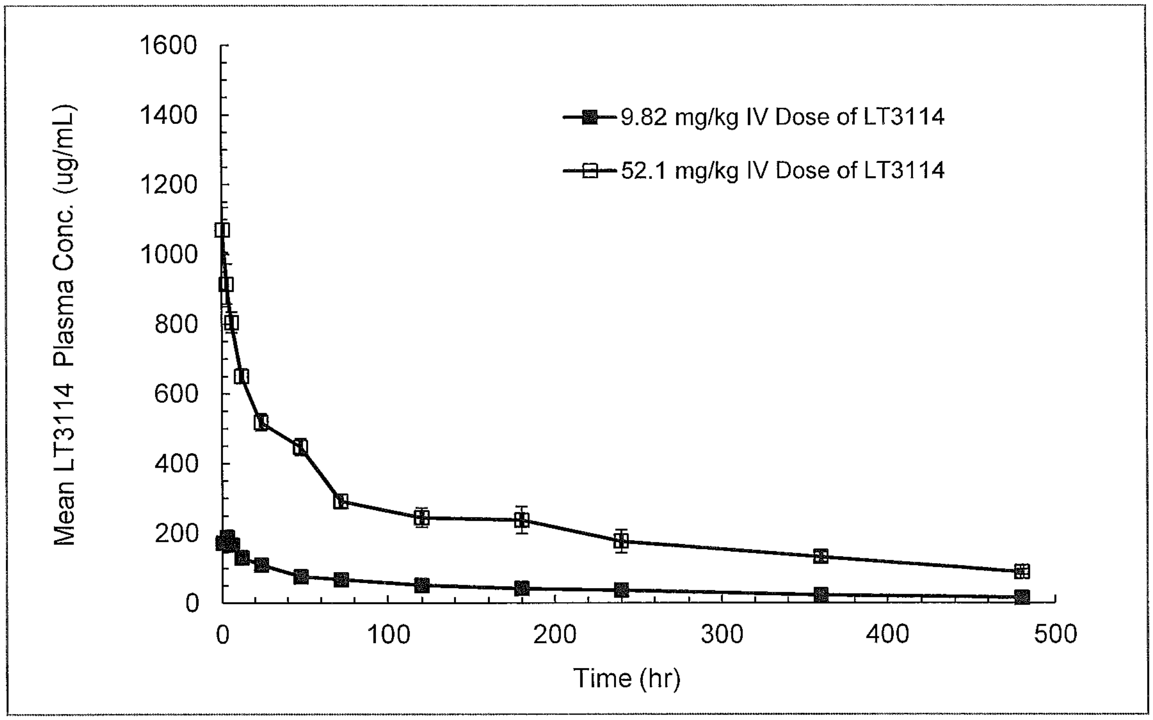
X. Figures

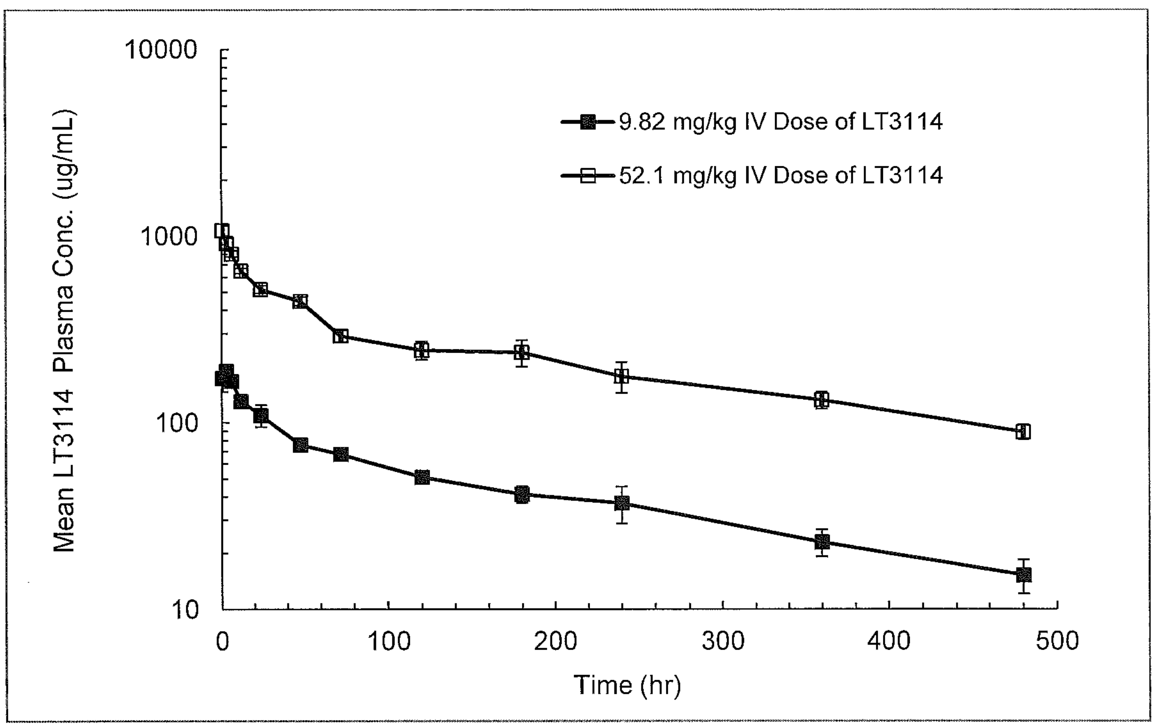
A. Figure 1 - Individual LT3114 Plasma Concentrations Plotted as a Function of Time Following a Single Intravenous Dose of LT3114 at ~10 mg/kg (Group 1) and ~50 mg/kg (Group 2) to Male Sprague-Dawley Rats (Rectilinear and Semi-Logarithmic Plots)





B. Figure 2 – Mean ± SD LT3114 Plasma Concentrations Plotted as a Function of Time Following a Single Intravenous Dose of LT3114 at ~10 mg/kg (Group 1) and ~50 mg/kg (Group 2) to Male Sprague-Dawley Rats (Rectilinear and Semi-Logarithmic Plots)





XI. Appendices

A. Appendix I—Test Article Information

Lpath Inc.	Certificate of Analysis

TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing	REVISION NO. 1.0.0

Name	Part Number	Lot Number
Lpathomab-DS, 30 mg/mL	**TBD**	GA121459

Formulation Buffer	Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0	2 to 8°C

Production Date	Test Date	Retest Date	Intended Use
JUN 2014	JUL 2014	DEC 2014	GLP-conforming Toxicity Testing

Manufacturer	Manufacturer Part Number	Manufacturer Lot Number
Gallus-MO, St. Louis, MO	TBD	**Gallus-MO 103948**

Attribute	Test	Specification	Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2	5.1
	CGE (Non-reduced)	Major band at 150 kDa	Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	iCE	Comparable to RS	Conforms
		Minor Peak at pI ~8.3 Major Peak at pI ~8.4 Minor Peak at pI ~8.6 Faint Peak at pI ~8.7	15% 60% 13% 4%
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL	32.1 mg/mL
	LPA Binding	100% ±40% of RS	89%

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL[1]
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance: ______________________ **Date: 22AUG2014**

Acting Head of QA - Frieder K. Hofmann, PhD

[1] At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

B. Appendix II—Protocol and Protocol Deviations

CALVERT LABS
Customized. Responsive. **Proven.**

Study Protocol

Title:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)
Calvert Study No.:	0835RL35.001
Testing Facility:	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447
Study Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121

I. Table Of Contents

II. Introduction

A. Title

A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)

B. Objective

To evaluate the pharmacokinetics of LT3114 in male Sprague Dawley rats after a single intravenous dose of LT3114 at two different dose levels.

C. Regulatory Compliance

This is a non-regulated study. However, it will be run according to the Standard Operating Procedures (SOPs) of Calvert. There will be no formal involvement of Calvert Quality Assurance.

D. Calvert Study Number

0835RL35.001

E. Testing Facility

Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

F. Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

G. Study Director

Steven D. Sloneker, B.S.
Director, Pharmacokinetics and ADME
Calvert Laboratories, Inc.
Scott Technology Park
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Scott Township, PA 18447
Phone: (570) 585-2225
Fax: (570) 585-2383

E-mail: steven.sloneker@calvertlabs.com

H. Study Monitor

Marlene W. Modi, Ph.D.
MWM Consulting Group, Inc. for Lpath Inc.
14 Joseph Court
Monmouth Junction, NJ 08852
Phone: (732) 274-9562
Mobile: (732) 822-6101
E-mail: mwmconsultinggroup@gmail.com

I. Sponsor Representative

Gary Woodnutt, Ph.D.
Senior Vice President Research
Lpath Inc
4025 Sorrento Valley Blvd
San Diego, CA 92121

J. Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis

Victor J. Johnson, Ph.D.
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Email: vjohnson@brt-labs.com

K. Key Study Dates

Proposed First Day of Dosing:	17 Jul 2014
Proposed Experimental Completion Date (In-Life Portion):	1 Aug 2014

III. Materials and Methods

A. Test Article and Test Article Formulations

1. Test Article

Identification: LT3114

Lot/Batch No.:	103948
Physical Description:	Clear colorless liquid
Storage Conditions:	Stored refrigerated (2-8°C) and protected from light.

2. Dose Preparations

Before dose preparation, 2 ml of a 10% polysorbate 80 solution (provided by the Sponsor; Batch# 104012-001, MM#306191), will be added aseptically to each liter of formulation buffer (also supplied by the Sponsor and in sterile containers; Batch# 104011-001, MM#306190), and mixed gently to ensure distribution throughout the volume of buffer. This diluent will then be used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations. The Sponsor will provide the test article already formulated at 32.1 mg/ml in sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared. Do not shake the formulation. Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

3. Dose Formulation Samples

Triplicate 1-ml samples of each dosing solution will be obtained. These samples will be stored at refrigerated temperatures (2-8°C) until shipment to the Sponsor-designee.

4. Formulated Test Article Analysis

Dose formulation samples will be shipped to and analytical evaluation will be done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. Concentration of LT3114 will be measured.

5. Accountability and Disposition

Unused test article will be returned to the Sponsor or designee at the completion of this study or, if necessary, retained for use on related future studies. The Sponsor will be notified in advance of shipping and a transmittal letter will accompany the shipment. The material will be packed in a suitable container to maintain the conditions specified by the Sponsor during transit plus an adequate margin of safety to account for any possible transit delays. If material is returned, the Sponsor will provide shipping instructions.

B. Test System (Animals and Animal Care)

1. *Description*

Species:	Rat
Stock:	Sprague Dawley Crl:CD(SD)
Total Number:	8; each animal will be surgically implanted with a catheter in a jugular vein using polyurethane tubing and glycerol or dextrose solution as the lumen lock solution.
Gender:	Male
Age Range:	8 to 14 weeks at start of dosing; records of dates of birth for animals used in this study will be retained in the Calvert archives.
Body Weight Range:	275 to 325 g
Animal Source:	Charles River Laboratories, Inc.
Experimental History:	Purpose-bred and experimentally naïve at the outset of the study
Identification:	Ear tag and cage card

2. *Rationale for Choice of Species and Number of Animals*

The Sprague Dawley rat was chosen for this study as it is a species that is used for non-clinical toxicity and pharmacokinetic evaluations and satisfies the regulatory requirement for non-clinical safety studies in rodents (1). Previous research has been conducted with the test article in rats. The test article, a human anti-LPA antibody (LT3114) and the murine parental molecule have both been evaluated for efficacy in a variety of rat and mouse animal models of disease. In models using Sprague Dawley rats efficacy has been examined in the STZ induced model of diabetic neuropathy, a chemotherapeutic- induced model of peripheral neuropathy and in models of acute inflammatory pain (carrageen or Freunds adjuvant induced) with doses (intravenous or subcutaneous) ranging from 10 to 40 mg/kg.

In addition single and multiple doses pharmacokinetic studies have been conducted with LT 3114 in murine models at dose levels ranging from 10 to 30 mg/kg. Four animals per dose group will allow the use of descriptive statistical analysis of the data.

3. *Husbandry*

Housing:	Animals will be individually housed in plastic totes with bedding upon receipt in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals will be kept will be documented in the study records. No other species will be kept in the same room.
Lighting:	12 hours light/12 hours dark except when room lights will be turned on during the dark cycle due to in-life procedures.
Room Temperature:	20 to 26 °C
Relative Humidity:	30 to 70%
Food:	All animals will have access to Teklad Certified Rodent Diet or equivalent *ad libitum*. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which will be maintained in the Calvert archives.
Water:	Water will be available *ad libitum* to each animal. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis will be maintained in the Calvert archives.
Acclimation:	Study animals will be acclimated to their laboratory environment for a minimum of one day prior to their dosing.

4. *Prestudy Health Screen and Selection Criteria*

All animals received for this study will be assessed as to their general health by a member of the veterinary staff or other authorized personnel. During the acclimation period, each animal will be observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that are determined to be suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

5. Assignment to Study Groups

Animals will be manually assigned to the study based on general health, body weight and patent catheter.

6. Humane Care of Animals

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge, this study is not unnecessary or duplicative.

C. Dose Administration

1. Treatments and Dosing Details

Trt	# of Animals and Sex	Route of Administration	Dose Conc. (mg/ml)	Dose Volume (ml/kg)	Dose Level (mg/kg)
1	4 males	IV	2	5	10
2	4 males	IV	10	5	50

2. Dosing

Route: Intravenous

Frequency: Single intravenous injection

Procedure: The intravenous formulation will be administered as a manual slow bolus injection over a period of approximately 3 minutes into a lateral tail vein.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans. Dose levels were selected by the Sponsor. The anticipated lowest pharmacologically active dose is 30 mg/kg.

D. In-Life Observations and Measurements

1. Clinical Observations

Frequency: Animals will be observed at least once daily and at all times of blood sampling. Any abnormal clinical signs will be recorded.

2. Body Weight

Frequency: Animals will be weighed on the day of test article administration prior to dosing.

E. Pharmacokinetics

1. Dosing Syringes

The actual dose administered to the animal will be the difference between the loaded and unloaded dosing syringe weights. Syringe weights will be measured and recorded to at least two decimal places.

2. Blood Sampling

Serial blood samples (~0.25 mL each), will be obtained from each animal via the jugular vein catheter at the following timepoints post-dose: 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours. Just before each blood sampling timepoint, blood will be withdrawn (a volume slightly larger than the catheter tubing void volume) and discarded. A blood sample (approximately 0.25 mL) will be obtained from the jugular vein catheter using a syringe with a blunted needle and then the blood sample will be immediately transferred into a blood collection tube. Following blood sample collection at 0.5, 3, and 6 hours post-dose, the catheter will be flushed with sterile saline. Following blood sample collection at 12, 24, 48, 72, 120, 180, 240, and 360 hours post-dose, the catheter will be flushed with sterile saline and then locked with a locking solution. The flush volume will be slightly larger than the catheter tubing void volume. Collected blood samples will be gently inverted several times and stored on chill packs. Blood samples will be collected into chilled tubes containing K_2EDTA as the anti-coagulant. Blood tubes will be placed on chill packs until being centrifuged within 30 minutes of collection.

3. Plasma Harvesting

Plasma will be harvested following centrifugation of the chilled blood samples at 3000 RPM for 10 minutes at +4 °C. Derived plasma will be transferred into prelabeled plastic vials (label designating Calvert study number, animal number, route, dose level/test article, date of collection, time of collection, and sample type) and stored frozen at -70 °C or lower.

4. Sample Shipment

Frozen plasma samples will be shipped frozen on dry ice via courier overnight delivery to and bioanalytical evaluation will be done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560.

A cover letter, sample transmittal form, and sample log will be sent with the samples.

F. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

All surviving animals will be euthanized by CO_2 asphyxiation following their final blood sampling.

b) Unscheduled Sacrifice/Gross Necropsy

Moribund animals may be euthanized by CO_2 asphyxiation for humane reasons, following consultation with a veterinarian, and, if possible, with the Sponsor. Any animal found dead or euthanized in a moribund condition will be subjected to a gross pathological exam at additional expense to the Sponsor. Any observations will be noted and any gross lesions will be preserved in Formalin for potential future analysis. Following necropsy and collection of any gross lesions the carcass will be discarded. Otherwise, no gross necropsy is scheduled to be conducted.

2. Gross Necropsy

Any animal found dead or euthanized in a moribund condition will be subjected to a gross pathological exam at additional expense to the Sponsor. Any observations will be noted and any gross lesions will be preserved in Formalin for potential future analysis. Following necropsy and collection of any gross lesions the carcass will be discarded.

IV. Records and Reports

A. Data Analysis

Data analysis will consist of descriptive statistical analysis of the animal body weights and dosing details per treatment. Animal observations and blood sampling times (actual and elapsed) will be recorded in the raw data. Bioanalysis of the plasma samples will be the responsibility of Burleson Research Technologies (Morrisville, NC 27560). Pharmacokinetic analysis of the plasma sample assay results will be the responsibility of Calvert. Using noncompartmental analysis and WinNonlin Version 5.3 software, individual pharmacokinetics parameters will include $AUC_{0\text{-}inf}$, $AUC_{0\text{-}last}$, C_{max}, Cl, C_{last}, C_0, T_{last}, T_{max}, $T_{1/2}$, V_{ss}, and V_c. Note: If the terminal half-life cannot be reliably calculated, the dependent PK parameters of $AUC_{0\text{-}inf}$, Cl, V_{ss}, and V_c will not be reported.

B. Storage of Records

Test article tracking, in-life data, protocol, protocol amendments (if applicable), technical notes (if applicable), and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. Six months following the submission of the draft report, if there are no client comments generated by the Sponsor and/or Study Monitor, the Sponsor/Study Monitor will be notified and the report will be finalized and archived according to the terms stated in the protocol. After 2 years following report finalization, the Sponsor will be contacted to determine final disposition of all study materials.

V. Miscellaneous

A. Confidentiality Statement

The information contained herein is for the personal use of the intended recipient(s).

B. References

1. Animal Models in Toxicology. Second Edition. Edited by Shayne C. Gad. Chapter 3 – The Rat. *Metabolism*. Pages 217-275.

VI. Protocol Approval Signatures

Steven D. Sloneker | 27 Jun 2014

Study Director's Signature | Date

Steven Sloneker, B.S.

Calvert Laboratories, Inc.

27 JUN 2014

Scientific Management | Date

Calvert Laboratories, Inc.

Terri L. Gyza | 26 June 2014

IACUC | Date

Calvert Laboratories, Inc.

Marlene Modi | June 26, 2014

Marlene Modi, Ph.D. | Date

Study Monitor

Gary Woodnutt, Ph.D. | Date

Sponsor Representative

VI. Protocol Approval Signatures

Study Director's Signature | Date
Steven Sloneker, B.S.
Calvert Laboratories, Inc.

Scientific Management | Date
Calvert Laboratories, Inc.

IACUC | Date
Calvert Laboratories, Inc.

Marlene Modi, Ph.D. | Date
Study Monitor

26th June 2014

Gary Woodnutt, Ph.D. | Date
Sponsor Representative

Protocol Deviations

1. The protocol-specified animal body weight range was 275 to 325 g. The actual body weight range was 309 to 331 g. This slight deviation in the body weight range was not considered to have an impact on the outcome or integrity of the study since the animals were within the specified age range and the dose administered was normalized for body weight.

2. The protocol required the temperature and relative humidity ranges in the animal room to be 20 to 26°C and 30 to 70%, respectively. The ranges were 17 to 24°C and 30 to 92%, respectively. These deviations were not of a magnitude or duration to have had an impact on the study since the animals' health was not affected.

C. Appendix III—Dosing and Blood Sampling Times (Actual and Elapsed) and Clinical Observations

Rat #	Group #	Sex	Date	Dose Level (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1343	1	M	21-Jul-14	10.1	IV	8:00:15	8:03:15	0.5	8:30:24	0:30:09	Normal	8:56	9:20
1343	1	M	21-Jul-14	10.1	IV			3	11:01:54	3:01:39	Normal	11:15	11:50
1343	1	M	21-Jul-14	10.1	IV			6	14:01:48	6:01:33	Normal	14:25	14:50
1343	1	M	21-Jul-14	10.1	IV			12	20:02:11	12:01:56	Normal	20:25	20:50
1343	1	M	22-Jul-14	10.1	IV			24	8:00:18	24:00:03	Normal	8:25	8:50
1343	1	M	23-Jul-14	10.1	IV			48	7:59:54	47:59:39	Normal	8:25	8:50
1343	1	M	24-Jul-14	10.1	IV			72	7:59:51	71:59:36	Normal	8:25	8:50
1343	1	M	26-Jul-14	10.1	IV			120	8:01:49	120:01:34	Normal	8:25	8:50
1343	1	M	28-Jul-14	10.1	IV			180	20:00:00	179:59:45	Normal	20:25	20:50
1343	1	M	31-Jul-14	10.1	IV			240	8:02:40	240:02:25	Normal	8:25	9:00
1343	1	M	5-Aug-14	10.1	IV			360	8:02:48	360:02:33	Normal	8:25	8:55
1343	1	M	10-Aug-14	10.1	IV			480	8:00:05	479:59:50	Normal	8:25	8:55
1344	1	M	21-Jul-14	9.75	IV	8:05:01	8:08:13	0.5	8:34:54	0:29:53	Normal	8:56	9:20
1344	1	M	21-Jul-14	9.75	IV			3	11:05:50	3:00:49	Normal	11:15	11:50
1344	1	M	21-Jul-14	9.75	IV			6	14:05:04	6:00:03	Normal	14:25	14:50
1344	1	M	21-Jul-14	9.75	IV			12	20:05:48	12:00:47	Normal	20:25	20:50
1344	1	M	22-Jul-14	9.75	IV			24	8:04:54	23:59:53	Normal	8:25	8:50
1344	1	M	23-Jul-14	9.75	IV			48	8:04:48	47:59:47	Normal	8:25	8:50
1344	1	M	24-Jul-14	9.75	IV			72	8:04:46	71:59:45	Normal	8:25	8:50
1344	1	M	26-Jul-14	9.75	IV			120	8:06:32	120:01:31	Normal	8:25	8:50
1344	1	M	28-Jul-14	9.75	IV			180	20:04:41	179:59:40	Normal	20:25	20:50
1344	1	M	31-Jul-14	9.75	IV			240	8:07:24	240:02:23	Normal	8:25	9:00
1344	1	M	5-Aug-14	9.75	IV			360	8:07:40	360:02:39	Normal	8:25	8:55
1344	1	M	10-Aug-14	9.75	IV			480	8:04:51	479:59:50	Normal	8:25	8:55

Rat #	Group #	Sex	Date	Dose Level (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1345	1	M	21-Jul-14	9.66	IV	8:10:00	8:13:02	0.5	8:40:02	0:30:02	Normal	8:56	9:20
1345	1	M	21-Jul-14	9.66	IV			3	11:09:49	2:59:49	Normal	11:15	11:50
1345	1	M	21-Jul-14	9.66	IV			6	14:10:02	6:00:02	Normal	14:25	14:50
1345	1	M	21-Jul-14	9.66	IV			12	20:09:48	11:59:48	Normal	20:25	20:50
1345	1	M	22-Jul-14	9.66	IV			24	8:09:48	23:59:48	Normal	8:25	8:50
1345	1	M	23-Jul-14	9.66	IV			48	8:09:41	47:59:41	Normal	8:25	8:50
1345	1	M	24-Jul-14	9.66	IV			72	8:09:41	71:59:41	Normal	8:25	8:50
1345	1	M	26-Jul-14	9.66	IV			120	8:11:02	120:01:02	Normal	8:25	8:50
1345	1	M	28-Jul-14	9.66	IV			180	20:09:36	179:59:36	Normal	20:25	20:50
1345	1	M	31-Jul-14	9.66	IV			240	8:12:02	240:02:02	Normal	8:25	9:00
1345	1	M	5-Aug-14	9.66	IV			360	8:12:15	360:02:15	Normal	8:25	8:55
1345	1	M	10-Aug-14	9.66	IV			480	8:09:47	479:59:47	Normal	8:25	8:55
1346	1	M	21-Jul-14	9.71	IV	8:15:02	8:18:02	0.5	8:45:00	0:29:58	Normal	8:56	9:20
1346	1	M	21-Jul-14	9.71	IV			3	11:14:44	2:59:42	Normal	11:15	11:50
1346	1	M	21-Jul-14	9.71	IV			6	14:14:41	5:59:39	Normal	14:25	14:50
1346	1	M	21-Jul-14	9.71	IV			12	20:14:41	11:59:39	Normal	20:25	20:50
1346	1	M	22-Jul-14	9.71	IV			24	8:14:44	23:59:42	Normal	8:25	8:50
1346	1	M	23-Jul-14	9.71	IV			48	8:14:35	47:59:33	Normal	8:25	8:50
1346	1	M	24-Jul-14	9.71	IV			72	8:14:36	71:59:34	Normal	8:25	8:50
1346	1	M	26-Jul-14	9.71	IV			120	8:15:47	120:00:45	Normal	8:25	8:50
1346	1	M	28-Jul-14	9.71	IV			180	20:14:28	179:59:26	Normal	20:25	20:50
1346	1	M	31-Jul-14	9.71	IV			240	8:16:38	240:01:36	Normal	8:25	9:00
1346	1	M	5-Aug-14	9.71	IV			360	8:17:00	360:01:58	Normal	8:25	8:55
1346	1	M	10-Aug-14	9.71	IV			480	8:14:44	479:59:42	Normal	8:25	8:55

Rat #	Group #	Sex	Date	Dose Level (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1347	2	M	21-Jul-14	51.7	IV	8:20:03	8:23:04	0.5	8:49:45	0:29:42	Normal	8:56	9:20
1347	2	M	21-Jul-14	51.7	IV			3	11:19:40	2:59:37	Normal	11:35	11:50
1347	2	M	21-Jul-14	51.7	IV			6	14:19:36	5:59:33	Normal	14:25	14:50
1347	2	M	21-Jul-14	51.7	IV			12	20:19:35	11:59:32	Normal	20:25	20:50
1347	2	M	22-Jul-14	51.7	IV			24	8:19:40	23:59:37	Normal	8:25	8:50
1347	2	M	23-Jul-14	51.7	IV			48	8:19:30	47:59:27	Normal	8:25	8:50
1347	2	M	24-Jul-14	51.7	IV			72	8:19:32	71:59:29	Normal	8:25	8:50
1347	2	M	26-Jul-14	51.7	IV			120	8:19:42	119:59:39	Normal	8:25	8:50
1347	2	M	28-Jul-14	51.7	IV			180	20:19:21	179:59:18	Normal	20:25	20:50
1347	2	M	31-Jul-14	51.7	IV			240	8:20:40	240:00:37	Normal	8:25	9:00
1347	2	M	5-Aug-14	51.7	IV			360	8:21:31	360:01:28	Normal	8:25	8:55
1347	2	M	10-Aug-14	51.7	IV			480	8:19:41	479:59:38	Normal	8:25	8:55
1348	2	M	21-Jul-14	53.4	IV	8:25:04	8:28:06	0.5	8:54:34	0:29:30	Normal	8:56	9:20
1348	2	M	21-Jul-14	53.4	IV			3	11:24:31	2:59:27	Normal	11:35	11:50
1348	2	M	21-Jul-14	53.4	IV			6	14:24:19	5:59:15	Normal	14:25	14:50
1348	2	M	21-Jul-14	53.4	IV			12	20:24:27	11:59:23	Normal	20:25	20:50
1348	2	M	22-Jul-14	53.4	IV			24	8:24:30	23:59:26	Normal	8:25	8:50
1348	2	M	23-Jul-14	53.4	IV			48	8:24:26	47:59:22	Normal	8:25	8:50
1348	2	M	24-Jul-14	53.4	IV			72	8:24:28	71:59:24	Normal	8:25	8:50
1348	2	M	26-Jul-14	53.4	IV			120	8:24:36	119:59:32	Normal	8:25	8:50
1348	2	M	28-Jul-14	53.4	IV			180	20:24:17	179:59:13	Normal	20:25	20:50
1348	2	M	31-Jul-14	53.4	IV			240	8:24:51	239:59:47	Normal	8:25	9:00
1348	2	M	5-Aug-14	53.4	IV			360	8:26:12	360:01:08	Normal	8:25	8:55
1348	2	M	10-Aug-14	53.4	IV			480	8:24:36	479:59:32	Normal	8:25	8:55

Rat #	Group #	Sex	Date	Dose Level (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1349	2	M	21-Jul-14	52.0	IV	8:30:00	8:33:01	0.5	9:00:37	0:30:37	Normal	9:05	9:20
1349	2	M	21-Jul-14	52.0	IV			3	11:29:20	2:59:20	Normal	11:35	11:50
1349	2	M	21-Jul-14	52.0	IV			6	14:29:30	5:59:30	Normal	14:36	14:50
1349	2	M	21-Jul-14	52.0	IV			12	20:29:20	11:59:20	Normal	20:36	20:50
1349	2	M	22-Jul-14	52.0	IV			24	8:29:24	23:59:24	Normal	8:36	8:50
1349	2	M	23-Jul-14	52.0	IV			48	8:29:20	47:59:20	Normal	8:36	8:50
1349	2	M	24-Jul-14	52.0	IV			72	8:29:21	71:59:21	Normal	8:36	8:50
1349	2	M	26-Jul-14	52.0	IV			120	8:29:29	119:59:29	Normal	8:36	8:50
1349	2	M	28-Jul-14	52.0	IV			180	20:29:24	179:59:24	Normal	20:36	20:50
1349	2	M	31-Jul-14	52.0	IV			240	8:29:30	239:59:30	Normal	8:36	9:00
1349	2	M	5-Aug-14	52.0	IV			360	8:31:03	360:01:03	Normal	8:35	8:55
1349	2	M	10-Aug-14	52.0	IV			480	8:29:31	479:59:31	Normal	8:36	8:55
1350	2	M	21-Jul-14	51.5	IV	8:35:18	8:35:27	0.5	9:04:31	0:29:13	Normal	9:05	9:20
1350	2	M	21-Jul-14	51.5	IV			3	11:34:11	2:58:53	Normal	11:35	11:50
1350	2	M	21-Jul-14	51.5	IV			6	14:34:11	5:58:53	Normal	14:36	14:50
1350	2	M	21-Jul-14	51.5	IV			12	20:34:12	11:58:54	Normal	20:36	20:50
1350	2	M	22-Jul-14	51.5	IV			24	8:34:11	23:58:53	Soft Feces	8:36	8:50
1350	2	M	23-Jul-14	51.5	IV			48	8:34:13	47:58:55	Soft Feces	8:36	8:50
1350	2	M	24-Jul-14	51.5	IV			72	8:34:14	71:58:56	Soft Feces	8:36	8:50
1350	2	M	26-Jul-14	51.5	IV			120	8:34:20	119:59:02	Soft Feces	8:36	8:50
1350	2	M	28-Jul-14	51.5	IV			180	20:34:11	179:58:53	Normal	20:36	20:50
1350	2	M	31-Jul-14	51.5	IV			240	8:34:15	239:58:57	Normal	8:36	9:00
1350	2	M	5-Aug-14	51.5	IV			360	8:34:41	359:59:23	Normal	8:35	8:55
1350	2	M	10-Aug-14	51.5	IV			480	8:24:11	479:48:53	Normal	8:36	8:55

D. Appendix IV—Individual LT3114 Rat Plasma Concentration-Time Data

Group #	Rat #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (µg/mL)
1	1343	M	10.1	0.5	132.8
1	1343	M	10.1	3	208.5
1	1343	M	10.1	6	170.7
1	1343	M	10.1	12	121.3
1	1343	M	10.1	24	113.5
1	1343	M	10.1	48	73.2
1	1343	M	10.1	72	69.8
1	1343	M	10.1	120	50.5
1	1343	M	10.1	180	36.1
1	1343	M	10.1	240	26.1
1	1343	M	10.1	360	17.5
1	1343	M	10.1	480	10.6
1	1344	M	9.75	0.5	182.0
1	1344	M	9.75	3	172.1
1	1344	M	9.75	6	163.8
1	1344	M	9.75	12	139.2
1	1344	M	9.75	24	124.9
1	1344	M	9.75	48	77.5
1	1344	M	9.75	72	68.1
1	1344	M	9.75	120	52.4
1	1344	M	9.75	180	42.1
1	1344	M	9.75	240	43.2
1	1344	M	9.75	360	26.3
1	1344	M	9.75	480	16.7
1	1345	M	9.66	0.5	190.2
1	1345	M	9.66	3	185.5
1	1345	M	9.66	6	176.3
1	1345	M	9.66	12	133.8
1	1345	M	9.66	24	110.7
1	1345	M	9.66	48	76.6
1	1345	M	9.66	72	67.2
1	1345	M	9.66	120	47.9
1	1345	M	9.66	180	46.4
1	1345	M	9.66	240	43.7
1	1345	M	9.66	360	24.2
1	1345	M	9.66	480	16.2
1	1346	M	9.71	0.5	185.5
1	1346	M	9.71	3	190.8
1	1346	M	9.71	6	157.5
1	1346	M	9.71	12	126.4
1	1346	M	9.71	24	89.0
1	1346	M	9.71	48	76.6
1	1346	M	9.71	72	65.9
1	1346	M	9.71	120	53.5
1	1346	M	9.71	180	40.6
1	1346	M	9.71	240	35.0
1	1346	M	9.71	360	23.5
1	1346	M	9.71	480	17.5

Group #	Rat #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (µg/mL)
2	1347	M	51.7	0.5	1048.4
2	1347	M	51.7	3	930.9
2	1347	M	51.7	6	832.1
2	1347	M	51.7	12	626.3
2	1347	M	51.7	24	515.6
2	1347	M	51.7	48	478.0
2	1347	M	51.7	72	274.0
2	1347	M	51.7	120	224.2
2	1347	M	51.7	180	254.1
2	1347	M	51.7	240	170.7
2	1347	M	51.7	360	133.3
2	1347	M	51.7	480	92.5
2	1348	M	53.4	0.5	1165.8
2	1348	M	53.4	3	966.4
2	1348	M	53.4	6	778.4
2	1348	M	53.4	12	640.2
2	1348	M	53.4	24	541.7
2	1348	M	53.4	48	432.3
2	1348	M	53.4	72	306.6
2	1348	M	53.4	120	224.2
2	1348	M	53.4	180	281.1
2	1348	M	53.4	240	225.0
2	1348	M	53.4	360	138.8
2	1348	M	53.4	480	96.0
2	1349	M	52.0	0.5	1037.6
2	1349	M	52.0	3	832.1
2	1349	M	52.0	6	779.6
2	1349	M	52.0	12	672.2
2	1349	M	52.0	24	485.2
2	1349	M	52.0	48	424.4
2	1349	M	52.0	72	274.0
2	1349	M	52.0	120	247.2
2	1349	M	52.0	180	190.8
2	1349	M	52.0	240	151.6
2	1349	M	52.0	360	112.4
2	1349	M	52.0	480	77.5
2	1350	M	51.5	0.5	1026.9
2	1350	M	51.5	3	929.5
2	1350	M	51.5	6	827.9
2	1350	M	51.5	12	663.2
2	1350	M	51.5	24	529.7
2	1350	M	51.5	48	454.5
2	1350	M	51.5	72	312.3
2	1350	M	51.5	120	282.3
2	1350	M	51.5	180	224.8
2	1350	M	51.5	240	159.6
2	1350	M	51.5	360	143.4
2	1350	M	51.5	480	90.6

E. Appendix V—Analytical/Bioanalytical Subreport

STUDY PHASE REPORT

STUDY TITLE:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)
REPORT TITLE:	Determination of LT3114 in Dose Formulation Samples and Rat Plasma Pharmacokinetic Samples
BRT Reference No.:	BRT 20140725
Author:	Victor J. Johnson, PhD (BRT)
Dose Formulation and PK Analysis Laboratory:	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Testing Facility:	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447
Study Number:	0835RL35.001
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121

Report Date: 04 December 2014

COMPLIANCE – NON-GLP STUDY

Although this study was performed as indicated in the study protocol and applicable BRT standard operating procedures (SOPs), it was investigational in nature and was not expected to conform to good laboratory practice (GLP) standards.

APPROVALS

Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.

04 Dec 14
Date

Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

04 Dec 2014
Date

Report Review:

Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

04 Dec 2014
Date

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STUDY INFORMATION

Determination of LT3114 in Dose Formulation Samples and Rat Plasma Pharmacokinetic Samples

Study Title:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)
Testing Facility:	Calvert Laboratories, Inc.
Testing Facility Study No:	0835RL35.001
Study Director:	Steven D. Sloneker, B.S.
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121
Test Site (Dose Formulation Analysis and Plasma PK Analysis):	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Test Site Reference No.:	BRT 20140725
Principal Investigator:	Victor J. Johnson, PhD (BRT)
Analysts:	Kristen Fisher (BRT) Kim Barfield (BRT)
Study Initiation Date:	27 Jun 14
Sample Receipt Date:	12 Aug 14
Experimental Start Date:	19 Aug 14
Experimental Completion Date:	09 Sep 14
Study Phase Completion Date:	04 Dec 14

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples and rat plasma pharmacokinetic (PK) samples generated by Calvert Study No. 0835RL35.001: “A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)”.

Dose formulation samples were provided for two dose concentrations, 2 mg/mL and 10 mg/mL. The LT3114 concentrations determined for these samples were 2.0 mg/mL and 13.9 mg/mL, respectively. The result for the 2 mg/mL dose formulation samples met acceptance criteria for accuracy and precision. The result for the 10 mg/mL dose formulation samples met acceptance criteria precision but did not demonstrate acceptable accuracy (within 20% of nominal).

LT3114 was measurable in all post-dose rat plasma samples and ranged from 10.6 to 208.5 μg/mL and 77.5 to 1165.8 μg/mL in animals dosed with 10 mg/kg or 50 mg/kg, respectively. Pre-dose plasma samples were not collected per study protocol.

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples and rat plasma pharmacokinetic (PK) samples in support of the study entitled "A Pharmacokinetic Study of LT3114 Administered Intravenously in Rats (non-GLP)".

STUDY DESIGN

Dose Formulation Samples

Dose formulations (2 and 10 mg/mL) were prepared by Calvert on 21 Jul 14 according to the study protocol using LT3114 in a citrate buffer formulation supplied by the Sponsor at 32.1 mg/mL. Triplicate 1 mL samples were collected on 21 Jul 14 from the middle of each dosing solution and were stored at 2-8°C until they were shipped to BRT (Table 1). Dose formulation samples were received at BRT on 12 Aug 14 and logged into BRT inventory and stored at 2-8°C until analyzed for LT3114 concentration. Samples were returned to 2-8°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received on 12 Aug 14 at BRT.

Date of Collection	Group #	Description
21 Jul 14	1	LT3114 – Treatment#1 Dose Formulation @ 2 mg/mL
21 Jul 14	1	LT3114 – Treatment#1 Dose Formulation @ 2 mg/mL
21 Jul 14	1	LT3114 – Treatment#1 Dose Formulation @ 2 mg/mL
21 Jul 14	2	LT3114 – Treatment#2 Dose Formulation @ 10 mg/mL
21 Jul 14	2	LT3114 – Treatment#2 Dose Formulation @ 10 mg/mL
21 Jul 14	2	LT3114 – Treatment#2 Dose Formulation @ 10 mg/mL

Rat Plasma Pharmacokinetic Samples

Male Sprague Dawley rats were administered LT3114 in a citrate buffer formulation at 10 or 50 mg/kg. Route of administration was intravenous injection as a manual slow bolus injection over a period of approximately 3 minutes via the lateral tail vein. Serial blood samples were obtained from the jugular vein of each animal at 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose. Blood was collected into chilled K_2EDTA tubes and maintained cold until plasma was isolated within 30 minutes by centrifugation at 3000 RPM for 10 minutes at 4°C. Plasma samples were stored at ≤-70°C until shipped to BRT on dry ice (Table 2). Plasma samples were received at BRT on 12 Aug 14 and logged into inventory at BRT and stored at ≤-70°C until analyzed for LT3114 concentration. Samples were returned to ≤-70°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 2: Pharmacokinetic rat plasma samples received on 12 Aug 14 at BRT.

Animal #	Sex	Time Point (HrPD[1])	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
1343	M	0.5	1	10	2	5
1343	M	3	1	10	2	5
1343	M	6	1	10	2	5
1343	M	12	1	10	2	5
1343	M	24	1	10	2	5
1343	M	48	1	10	2	5
1343	M	72	1	10	2	5
1343	M	120	1	10	2	5
1343	M	180	1	10	2	5
1343	M	240	1	10	2	5
1343	M	360	1	10	2	5
1343	M	480	1	10	2	5
1344	M	0.5	1	10	2	5
1344	M	3	1	10	2	5
1344	M	6	1	10	2	5
1344	M	12	1	10	2	5
1344	M	24	1	10	2	5
1344	M	48	1	10	2	5
1344	M	72	1	10	2	5
1344	M	120	1	10	2	5
1344	M	180	1	10	2	5
1344	M	240	1	10	2	5
1344	M	360	1	10	2	5
1344	M	480	1	10	2	5
1345	M	0.5	1	10	2	5
1345	M	3	1	10	2	5
1345	M	6	1	10	2	5
1345	M	12	1	10	2	5
1345	M	24	1	10	2	5
1345	M	48	1	10	2	5
1345	M	72	1	10	2	5
1345	M	120	1	10	2	5
1345	M	180	1	10	2	5
1345	M	240	1	10	2	5
1345	M	360	1	10	2	5
1345	M	480	1	10	2	5
1346	M	0.5	1	10	2	5
1346	M	3	1	10	2	5
1346	M	6	1	10	2	5
1346	M	12	1	10	2	5
1346	M	24	1	10	2	5

Animal #	Sex	Time Point (HrPD[1])	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
1346	M	48	1	10	2	5
1346	M	72	1	10	2	5
1346	M	120	1	10	2	5
1346	M	180	1	10	2	5
1346	M	240	1	10	2	5
1346	M	360	1	10	2	5
1346	M	480	1	10	2	5
1347	M	0.5	2	50	10	5
1347	M	3	2	50	10	5
1347	M	6	2	50	10	5
1347	M	12	2	50	10	5
1347	M	24	2	50	10	5
1347	M	48	2	50	10	5
1347	M	72	2	50	10	5
1347	M	120	2	50	10	5
1347	M	180	2	50	10	5
1347	M	240	2	50	10	5
1347	M	360	2	50	10	5
1347	M	480	2	50	10	5
1348	M	0.5	2	50	10	5
1348	M	3	2	50	10	5
1348	M	6	2	50	10	5
1348	M	12	2	50	10	5
1348	M	24	2	50	10	5
1348	M	48	2	50	10	5
1348	M	72	2	50	10	5
1348	M	120	2	50	10	5
1348	M	180	2	50	10	5
1348	M	240	2	50	10	5
1348	M	360	2	50	10	5
1348	M	480	2	50	10	5
1349	M	0.5	2	50	10	5
1349	M	3	2	50	10	5
1349	M	6	2	50	10	5
1349	M	12	2	50	10	5
1349	M	24	2	50	10	5
1349	M	48	2	50	10	5
1349	M	72	2	50	10	5
1349	M	120	2	50	10	5
1349	M	180	2	50	10	5
1349	M	240	2	50	10	5
1349	M	360	2	50	10	5
1349	M	480	2	50	10	5
1350	M	0.5	2	50	10	5
1350	M	3	2	50	10	5
1350	M	6	2	50	10	5
1350	M	12	2	50	10	5
1350	M	24	2	50	10	5
1350	M	48	2	50	10	5

Animal #	Sex	Time Point (HrPD[1])	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
1350	M	72	2	50	10	5
1350	M	120	2	50	10	5
1350	M	180	2	50	10	5
1350	M	240	2	50	10	5
1350	M	360	2	50	10	5
1350	M	480	2	50	10	5

[1]HrPD – Hours Post-Dose

METHODOLOGY

Dose Formulation and Plasma PK Samples

Dose formulation samples (2 dose levels with 3 aliquots each; 6 total samples) were received frozen on dry ice on 12 Aug 14. Upon receipt at BRT, all samples were stored at ≤-70°C. The samples were analyzed for LT3114 concentrations at BRT.

Rat plasma PK samples (2 dose levels with 4 animals/dose and single aliquots for 12 time points per animal; 96 total samples) were received frozen on dry ice on 12 Aug 14. Upon receipt at BRT, all samples were stored at ≤-70°C. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples and plasma PK samples were analyzed for LT3114 concentration using an ELISA method that was transferred to BRT from the Sponsor (BRT 20140420). The ELISA method is provided in Appendix 1.

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, dilipidized bovine serum (DBS) assay buffer, dose formulation samples, and plasma PK samples were added to the coated plates and incubated to allow for capture of the LT3114. A horse radish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data Analysis

Data analysis was performed using SoftMax®Pro (Molecular Devices) software. Calibration curves were generated by plotting the mean OD_{450} values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD_{450} values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from replicate wells, by the sample dilution factor. Results were reported to one decimal place.

Acceptance Criteria

Calibration Curve

- Determined by the total number of points (with no more than 2 points masked, and no 2 points from the same standard masked).
- Correlation Coefficient ≥ 0.9
- Precision (%CV) between replicates ≤ 20%
- Accuracy of each standard within 20% of nominal from 3.9-500 ng/mL.

Sample Acceptance

- Precision (%CV) between replicates ≤ 20%
- Accuracy within 20% of nominal (Dose formulation samples only)
- Samples with absorbance values outside the upper limit of quantification (ULOQ at 500 ng/mL) were retested at a higher dilution.
- Samples with absorbance values outside the lower limit of quantification (LLOQ at 5 ng/mL) were retested at a lower dilution, unless already tested at the minimum dilution (1:10).

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples of each dose formulation are provided in Table 3. The concentration of LT3114 in the 2 mg/mL dose formulation was 2.0 mg/mL (%CV for triplicate determination was 12.8) which met acceptance criteria for accuracy and precision. The concentration of LT3114 in the 10 mg/mL dose formulation was 13.9 mg/mL (%CV for triplicate determination was 5.6). The concentration of LT3114 in the

10 mg/mL dose formulation did not meet acceptance criteria for accuracy (>20% from nominal) but demonstrated acceptable precision among the replicates.

Rat plasma PK samples were analyzed for LT3114 concentrations (Table 4) All plasma samples provided results with acceptable precision. LT3114 was measurable in all post-dose rat plasma samples and ranged from 10.6 to 208.5 µg/mL and 77.5 to 1165.8 µg/mL in animals dosed with 10 mg/kg or 50 mg/kg, respectively. Pre-dose plasma samples were not collected per study protocol.

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 1 year from the date that the final report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

There were no deviations from the Study Protocol during the study phase.

TABLES

Table 3: LT3114 concentrations in dose formulation samples.

Batch Number	Dose Formulation	Samples	LT3114 (mg/mL)	% of Target	Mean LT3114 (mg/mL)	% of Target	%CV
21 July 14	LT3114 Low-Dose: 2mg/ml	Middle (1/3)	2.0	100	2.0	100	12.8
	LT3114 Low-Dose: 2mg/ml	Middle (2/3)	2.2	110			
	LT3114 Low-Dose: 2mg/ml	Middle (3/3)	1.7	85			
	LT3114 Mid-Dose: 10mg/ml	Middle (1/3)	13.0	130	13.9	139	5.8
	LT3114 Mid-Dose: 10mg/ml	Middle (2/3)	14.4	144			
	LT3114 Mid-Dose: 10mg/ml	Middle (3/3)	14.3	143			

Table 4: LT3114 concentrations in rat plasma samples.

Collection Time point	Dose (mg/kg/dose)	Group/Sex	Animal Number	LT3114 (μg/ml)
0.5h Post Dose (21Jul14)	10	1 Male	1343	132.8
		1 Male	1344	182.0
		1 Male	1345	190.2
		1 Male	1346	185.5
	50	2 Male	1347	1048.4
		2 Male	1348	1165.8
		2 Male	1349	1037.6
		2 Male	1350	1026.9
3h Post Dose (21Jul14)	10	1 Male	1343	208.5
		1 Male	1344	172.1
		1 Male	1345	185.5
		1 Male	1346	190.8
	50	2 Male	1347	930.9
		2 Male	1348	966.4
		2 Male	1349	832.1
		2 Male	1350	929.5
6h Post Dose (21Jul14)	10	1 Male	1343	170.7
		1 Male	1344	163.8
		1 Male	1345	176.3
		1 Male	1346	157.5
	50	2 Male	1347	832.1
		2 Male	1348	778.4
		2 Male	1349	779.6
		2 Male	1350	827.9
12h Post Dose (21Jul14)	10	1 Male	1343	121.3
		1 Male	1344	139.2
		1 Male	1345	133.8
		1 Male	1346	126.4
	50	2 Male	1347	626.3
		2 Male	1348	640.2
		2 Male	1349	672.2
		2 Male	1350	663.2
24h Post Dose (22Jul14)	10	1 Male	1343	113.5
		1 Male	1344	124.9
		1 Male	1345	110.7
		1 Male	1346	89.0
	50	2 Male	1347	515.6
		2 Male	1348	541.7
		2 Male	1349	485.2
		2 Male	1350	529.7
48h Post Dose (23Jul14)	10	1 Male	1343	73.2
		1 Male	1344	77.5
		1 Male	1345	76.6
		1 Male	1346	76.6
	50	2 Male	1347	478.0
		2 Male	1348	432.3
		2 Male	1349	424.4
		2 Male	1350	454.5

Collection Time point	Dose (mg/kg/dose)	Group/Sex	Animal Number	LT3114 (µg/ml)
72h Post Dose (24Jul14)	10	1 Male	1343	69.8
		1 Male	1344	68.1
		1 Male	1345	67.2
		1 Male	1346	65.9
	50	2 Male	1347	274.0
		2 Male	1348	306.6
		2 Male	1349	274.0
		2 Male	1350	312.3
120h Post Dose (26Jul14)	10	1 Male	1343	50.5
		1 Male	1344	52.4
		1 Male	1345	47.9
		1 Male	1346	53.5
	50	2 Male	1347	224.2
		2 Male	1348	224.2
		2 Male	1349	247.2
		2 Male	1350	282.3
180h Post Dose (26Jul14)	10	1 Male	1343	36.1
		1 Male	1344	42.1
		1 Male	1345	46.4
		1 Male	1346	40.6
	50	2 Male	1347	254.1
		2 Male	1348	281.1
		2 Male	1349	190.8
		2 Male	1350	224.8
240h Post Dose (28Jul14)	10	1 Male	1343	26.1
		1 Male	1344	43.2
		1 Male	1345	43.7
		1 Male	1346	35.0
	50	2 Male	1347	170.7
		2 Male	1348	225.0
		2 Male	1349	151.6
		2 Male	1350	159.6
360h Post Dose (05Aug14)	10	1 Male	1343	17.5
		1 Male	1344	26.3
		1 Male	1345	24.2
		1 Male	1346	23.5
	50	2 Male	1347	133.3
		2 Male	1348	138.8
		2 Male	1349	112.4
		2 Male	1350	143.4
480h Post Dose (10Aug14)	10	1 Male	1343	10.6
		1 Male	1344	16.7
		1 Male	1345	16.2
		1 Male	1346	17.5
	50	2 Male	1347	92.5
		2 Male	1348	96.0
		2 Male	1349	77.5
		2 Male	1350	90.6

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Rat Plasma
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS); assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H_2SO_4 (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)

    1.1 LPA-Conjugated-BSA coating material should be kept frozen until use.

    1.2 After dilution, LPA-Conjugated-BSA coating material should be stored at 2-8° C for up to 4 months

2. Peroxidase conjugated-anti-human IgG secondary antibody

    2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%)

2.2 Store at -20° C for up to 1 year

3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)

3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly

3.2 Store at 2-8°C for up to 3 months

4. Blocking Solution 1xPBS, 1% BSA l, 0.1% Tween 20

4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X)

4.2 Mix thoroughly and sterile filter

4.3 Store at 2-8°C for up to 1 month

ELISA Method

1. Coating Plates

1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at ≤-70°C. Thaw the aliquot at room temperature.

1.2 Dilute LPA-Conjugated-BSA coating material to 1 µg/ml in 0.05 M Carbonate Buffer.

1.3 Coat a 96-well high-binding microtiter plate(s) with 100 µL/well coating solution.

1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.

2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).

3. Blocking Plates

3.1 Wash coated plate(s) 4 times with wash solution.

3.2 Dispense 150 µL/well of blocking solution to each well and cover the plate(s) with plate sealers.

3.3 Incubate plate(s) at 37°C for 1 hour ± 5 minutes.

4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)

4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)

4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.

4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS to achieve desired range (0 – 4,000 ng/ml)

4.4 Wash plate(s) 4 times (300µl/well) with wash solution.

4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.

5 Add samples to appropriate wells

5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.

5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.

6 Add HRP-goat-anti-human secondary antibody solution.

6.1 Wash plate(s) 4 times (300µl/well) with wash solution.

6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody

6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.

6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.

6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.

6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.

7 Substrate Addition

7.1 Wash coated plate(s) 4 times with wash solution.

7.2 Dispense 100 µl **COLD** TMB solution to each well.

7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.

8 Stop Solution Addition

8.1 Dispense 100 µl 1.0 M H_2SO_4 solution to each well.

8.2 Measure absorbance at 450 nm on microplate reader